

Web-Based Access and Visualization of Hazardous Air Pollutants

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Running Title:

Visualization of Hazardous Air Pollutants

Key Words:

Cumulative Exposure Project, HAPs, Air Toxics, Environmental Data, Linked Micromap Plots, Micromaps, Graphics Production Library, Statistical Graphics, Statistical Summaries, World Wide Web.

Acknowledgement:

EPA funded the majority of the work behind this paper under contract No. 8W-1712-NTEX and cooperative agreement No. CR825564-01-0. Additional federal agencies, BLS and NCHS, supported some facets of this work. The article has not been subject to review by any of these agencies so does not necessarily reflect the view of the agencies, and no official endorsement should be inferred. The conclusions and opinions are solely those of the authors and are not necessarily the views of the agencies.

Abbreviations:

ASPEN:	Assessment System for Population Exposure Nationwide
BLS:	Bureau of Labor Statistics
CEP:	Cumulative Exposure Project
CGI:	Common Gateway Interface
EDF:	Environmental Defense Fund
EPA:	U.S. Environmental Protection Agency
GIS:	Geographic Information System
GPL:	Graphics Production Library
HAP:	Hazardous Air Pollutant
HEM:	Human Exposure Model
LM Plot:	Linked Micromap Plot
OECD:	Organization for Economic Cooperation and Development
PM:	Particulate Matter
TRI:	Toxics Release Inventory
VOC:	Volatile Organic Compound

Abstract:

This paper reports on a project that provides Web-based access to extensive hazardous air pollutant (HAP) model-based summaries from the U.S. Environmental Protection Agency (EPA). As part of EPA's Cumulative Exposure Project (CEP), long-term cumulative concentrations of 148 HAPs for the 60,803 census tracts in the 48 contiguous states have been modeled for 1990. The model results include estimates and confidence bounds that assess the estimated uncertainties for each of the HAPs in each census tract. The project challenge was to concisely display 148 x 60,803 estimates along with uncertainty bounds.

The project goal was to make these statistical summaries accessible to the public as statistical tables and graphs. The Web provides an easy way to make this information electronically accessible. One big challenge is to make the summaries conceptually accessible. The most difficult part of this is to communicate an understanding of the underlying data limitations, the modeling process, and a notion of how to interpret the model results. The easier part of conceptual accessibility addressed here concerns navigation through the summaries and consideration of values within a large context. Our context and focus approach provides interactive selection of a HAP coupled with a drill-down through the levels of a geo-political hierarchy. The hierarchy consists of states within the U.S., counties within states, and census tracts within counties.

Our Web-based approach also attended to the design of tables and graphics with the intent of making them more readable and useful. For tables, our approach focused on perceptual details such as rounding and foreground and background contrast. For graphics, our approach provided spatial context through the use of recently developed templates called linked micromap (LM) plots. Both, tables and micromaps, provide a hierarchically clickable drill-down to finer details. This provides fast answers to questions about the air quality in a given region in the contiguous United States.

1. Introduction

Over the last few years researchers have developed many improvements to make statistical graphics more accessible to the general public. These improvements include making statistical summaries more visual and providing more information at a time. Research in this area involved converting statistical tables into plots (1), new ways of displaying geographically referenced data (2), and, in particular, the development of linked micromap (LM) plots, often simply called micromaps (3-5). Another recent development is the JAVA-based Graphics Production Library (GPL) for the Web-based distribution of interactive statistical graphics (6). These graphics are applicable to the distribution of federal statistical summaries such as the description of hazardous air pollutants (HAPs).

Staff at the U.S. Environmental Protection Agency (EPA) and contractors modeled 1990 long-term cumulative concentrations for 148 HAPs at the census tract level. This resulted in 148 x 60,803 estimates along with upper and lower confidence bounds for each estimate. The main goal of our project was to provide Web-based access to this HAP data. We focused on providing a concise display with easy access to the data. Another goal was to make the model results easily understandable by a non-statistical audience. To achieve our goals, we developed interactive tables and extended the GPL by adding micromaps. The micromaps serve as a geographic navigational drill-down tool as well as being meaningful statistical overviews in their own right. The ability to drill down from the national overview showing states, to a state overview showing counties, to a county view showing census tracts provide rapid access to the fine grained detail in this substantial data set.

In Section 2 of this paper we describe EPA's Cumulative Exposure Project (CEP). We describe graphical statistical components, i.e., GPL and micromaps, that are used to construct the CEP Web site in Section 3. Section 4 provides deeper insights into the CEP Web site and the user's point of view. We finish with a discussion in Section 5. Additional details and a different set of micromap displays and screenshots from the CEP Web site can be found in Symanzik et al. (7).

2. EPA's Cumulative Exposure Project

Much of the characterization of air pollution has focused on air pollutants designated as "criteria pollutants" in the Clean Air Act, such as particulate matter, ozone and lead (8). This is largely due to the obvious health effects demonstrated by major pollution episodes, such as those in Donora, Pennsylvania, and London (9, 10), and the extensive availability of monitoring data to use in assessing health effects. Relatively little is known about the potential health effects of other air pollutants, a number of which are designated as "hazardous air pollutants" (HAPs) in the Clean Air Act. HAPs have been associated with a variety of adverse health outcomes, including cancer and noncancer effects, such as neurological, reproductive and developmental effects, mostly through occupational and animal studies (11).

Past analyses have relied on limited emissions and monitoring data and some modeling to assess public health impacts of air toxics. Some studies have attempted to assess differential impacts of air toxics on communities of color using emissions estimates, mostly from the Toxics Release Inventory (TRI), which contains emissions estimates from major manufacturers in the

United States (12, 13). Other analyses have attempted to characterize the potential public health impacts of air toxics (13-17). One set of studies uses monitoring data and concentrations estimated by dispersion modeling of emissions from a subset of commercial and industrial facilities to evaluate potential noncancer health effects (14-16). These studies found that outdoor concentrations were often greater than benchmarks representing thresholds for potential public health impacts. However, these studies were not comprehensive in their scope either due to lack of monitoring data (18) or emissions data.

A recent analysis by Woodruff et al., as part of the U.S. Environmental Protection Agency's (EPA) Cumulative Exposure Project (CEP), has assessed the potential public health implications of air toxics across the United States for 1990 (19). This analysis uses modeled outdoor concentrations of air toxics across the contiguous United States (20) to help address the lack of data on outdoor concentrations. Emissions data from stationary and mobile sources are used as inputs into a dispersion model which estimates 1990 average outdoor concentrations of 148 air toxics for every census tract in the contiguous United States. The estimated outdoor concentrations from the analysis are used as a reasonable proxy for potential exposure in making relative comparisons of hazard and performing screening level analysis. Comparison of the estimated air toxics to previously defined levels (benchmark concentrations) representing thresholds of concern for potential adverse public health impacts found that many estimated concentrations are above benchmark concentrations (19, 21).

2.1 Estimating 1990 Outdoor Concentrations of Hazardous Air Pollutants

Outdoor concentrations of hazardous air pollutants were estimated using a Gaussian dispersion model (20, 22). The Assessment System for Population Exposure Nationwide (ASPEN) used in this study is a modified version of EPA's Human Exposure Model (HEM) (22), a standard tool designed to model long-term concentrations over large spatial scales. Long-term average concentrations of HAPs were calculated at the census tract level based on emissions rates of the HAPs and frequencies of various meteorological conditions, including wind speed, wind direction, and atmospheric stability. In addition, the model used in this analysis incorporates simplified treatment of atmospheric processes such as decay, secondary formation, and deposition. There are 60,803 census tracts in the contiguous U.S. that vary in physical size but typically have 4,000 residents.

The pollutants chosen for modeling were based on the list of 189 HAPs in section 112 of the 1990 Clean Air Act Amendments. A baseline year of 1990 was selected for modeling. Available emissions data were reviewed and appropriate data were identified for 148 HAPs.

A national inventory of HAP emissions was developed for this study as a required input to the dispersion model. For large manufacturing sources, emissions data contained in EPA's Toxics Release Inventory were used (23). Emissions estimates were developed for other sources, such as large combustion sources, automobiles and dry cleaners, using HAP speciation data in combination with EPA's extensive national inventories of 1990 emissions of total volatile organic compounds (VOCs) and particulate matter (PM) (24, 25). HAP emissions were derived from VOC and PM emissions estimates by applying industry-specific and process-specific estimates of the presence of particular HAPs in a VOC or PM emissions stream (20). Alaska and Hawaii are not included in this study because the national VOC and PM emissions inventories do not include data for these states.

The dispersion model accounted for long-term concentrations of HAPs attributable to current (i.e., 1990) anthropogenic emissions within 50 kilometers of each census tract centroid. For 28 HAPs, estimated outdoor concentrations also included a “background” component attributable to long-range transport, re-suspension of historical emissions, and natural sources derived from measurements taken at “clean air” locations remote from the impact of local anthropogenic sources (20). Comparison of the 1990 modeled concentrations to available air toxics monitoring data for 1990 indicates that the model estimates are typically of the correct magnitude, with a general tendency to underestimate the measured ambient concentrations.

EPA's CEP Web site (<http://www.epa.gov/CumulativeExposure>) has been specifically designed to provide further insight into statistical methods and methodologies and answers questions related to air toxics. In addition to providing explanatory texts, documents, and external links that relate to the material described in this section, one of the main intentions of the CEP Web site was to provide information about the estimated air toxics data. A key piece of assessing the data is the ability to evaluate the information visually. The remainder of this paper describes the work done to present the data and the development of the CEP Web site.

3. Graphical Statistical Components

In this section, we want to introduce the main graphical statistical components that are used for the Web-based access and visualization of HAPs through EPA's CEP Web site.

3.1 The Graphics Production Library

The Graphics Production Library (GPL) is a JAVA class library of graphics routines that form the basis for a convenient creation and modification of statistical graphics on the Web (6, http://www.monumental.com/dan_rope/gpl/). The GPL was initially developed within the Bureau of Labor Statistics (BLS) to facilitate the Web-based distribution of BLS's statistical summaries. It provides interactive features such as dragging and dropping data columns atop of each other to allow easier comparisons of the data, reordering and rescaling of panels, and panning and zooming. Thus, it considerably extends the static features but otherwise closely follows the row-labeled plots of Carr (1). Recent recommendations on statistical graphics as given in Cleveland (26, 27) have also been followed during the design of the GPL. Moreover, the GPL provides access to meta data, i.e., the possibility to add links to articles associated with the data or to include warning flags within a data display. The GPL currently supports three types of graphics displays: bar plots, dot plots, and time series graphics. Other types of graphics have been planned but have not been implemented so far by the BLS. Unfortunately, the GPL does not provide any functionality to draw maps and link statistical data to these, however this is one of the features that has been planned.

3.2 Micromaps

Linked micromap (LM) plots, often simply called micromaps, provide a new statistical paradigm for the viewing of spatially referenced statistical summaries in the corresponding spatial context. Full details on LM plots can be found in Carr et al. (3-5). A LM plot for the 50 states within the U.S. provides an alternative to displaying all statistical information on a single

choropleth map. Instead, several small maps (10 maps in our 50-state example) are drawn. The associated statistical data is arranged according to a particular criterion (which might be from highest to lowest or with respect to the alphabetical ordering of the corresponding geographical region). Then, the five highest values of the data are drawn in a statistical plot (e.g., dot plot, bar plot, box plot, plot with confidence bounds, time series plot) on the right side of the first small map. For each data point a different color is used in the statistical plot. The corresponding regions (in this case five states) are highlighted in the same colors on the first small map. All other states remain uncolored in this map. The same is done for the next five highest remaining data points in the second small map and associated statistical display. We continue until all data points/regions have been plotted/highlighted.

This splitting into several maps provides an immediate overview where the high/middle/low observations can be found geographically. We can judge if there are any geographic clusters or if the underlying measurements are randomly spread over the area under consideration. LM plots can display multiple statistical variables at a time. Examples of micromaps and S-Plus code that has been written for their creation can be found at <ftp://galaxy.gmu.edu/pub/dcarr/newsletter/micromap/>.

Our example at the state level for Pennsylvania in Figure 1 that has been created using S-Plus generally follows the design principle described above. However, we had to find an (almost) symmetric display for 67 counties. We ended up with 16 maps, 13 of them with 4 counties and 3 of them with 5 counties. Moreover, the layout has been split into four quarters. In addition to simply highlighting the individual 4 or 5 counties from the associated statistical display in each map, all 16 or 17 counties that fall into the corresponding quarter are highlighted on this map. This allows a much faster understanding of the spatial structure. As an example, the viewer can immediately gasp that the highest benzene concentrations have been modeled for counties surrounding major cities (e.g., Philadelphia and Pittsburgh) while the lowest benzene concentrations have been modeled for counties along the sparsely populated border region to New York.

INSERT FIGURE 1 ON PREVIOUS OR NEXT PAGE

4. The CEP Web Site

4.1 The User's Point of View

In addition to providing access to explanatory texts and entire CEP-related documents, the main intention of the CEP Web site is to provide fast and easy access to the 148 HAPs at different spatial resolutions ranging from the (top) U.S. level to the (bottom) census tract level.

Three mechanisms have been designed to select main features of the CEP Web site and to maneuver from the U.S. to the census tract level and up again. Standard menus in the upper right part of the Web page allow the user to select the representation (Data Tables, Micromaps, or Raw Data), one of the 148 HAPs, a state and, based on this selection, a county for which the data should be displayed (see Figure 2). This navigation and selection menu remains permanently visible, independently from the current statistical display.

The second tool that has been implemented to drill down through the levels of a geopolitical hierarchy makes use of interactive tables that display statistical data and serve as

navigational tools at the same time. Figure 2 shows such a table at the state level for Pennsylvania. The user can mouse-click on any of the listed counties and a new table will appear, now displaying the census tract data (including a 90% confidence interval) for all census tracts within the selected county. In addition, the small arrows pointing upwards and downwards allow the user to rearrange the rows of the table increasingly or decreasingly according to the selected criterion. In the current view, the data is ordered from highest to lowest median benzene concentration. This is indicated through the larger downward arrow and the differently colored background of this data column. Through this interactivity, interesting information can be found in a larger table within seconds instead of having to search sequentially through each table column to find largest, smallest, and - even more complicated - center values.

INSERT FIGURE 2 ON PREVIOUS OR NEXT PAGE

It should be noted that, when designing our tabular display, we spend particular attention to recommendations from the cognitive sciences. Numbers have been rounded to two significant digits (with respect to the smallest number in any given display) and colors have been selected in such a way to produce a pleasant visual effect for the human observer. Also, icons have been incorporated that warn users of suspect numerical values. These are usually particular census tract/HAP combinations for which EPA assumes that the modeled 1990 concentrations are considerably overestimated (see Figure 3). In addition to rounded data, the raw data is available through this CEP Web site. This data format is most useful for users that want to conduct additional statistical analyses.

INSERT FIGURE 3 ON NEXT PAGE

The third navigational tool is based on so-called hierarchical clickable micromaps in the GPL environment. This approach combines LM plots and the GPL and extends their joint features to allow dynamic access to complex, geographically referenced data. The user can mouse-click on a region (state, county) in the map and the display changes with respect to the selected region. Thus, micromaps serve as a navigational tool but also each individual LM plot is a sophisticated statistical display by itself as described in Section 3.2. While the idea of using micromaps simultaneously as a statistical display and for navigational purposes first has been considered for EPA's CEP Web site, this methodology may be of benefit in innumerable future applications.

In the CEP Web site, after having selected one state in the top U.S. micromap (or tabular display), there will be a new micromap (or tabular display) - however, this time at the county level for the selected state. This time, the user can select a county and reaches the lowest level in this hierarchy - a graphical display (or tabular display) at the census tract level. This geographic selection is an easy way to maneuver through the 60,803 census tracts even for inexperienced users of the Web. At the higher levels (U.S., state), statistical summaries such as mean, median, minimum, maximum, and quartiles with respect to all census tracts in each region are displayed (see Figure 2). At the census tract level, uncertainty bounds are displayed (see Figure 3).

4.2 Creation of Micromaps

Before we can implement LM plots in the GPL environment, generalized maps that form the basis of micromaps have to be created. Previous LM plots within the literature (3-5) make use of hand-created generalized maps, e.g., of the United States or the OECD (Organization for Economic Cooperation and Development) states. The creation of such a generalized map by hand typically requires several hours of work - an option which is clearly not feasible if we are interested in generalized maps for all 50 states or even all counties within the United States.

We are not aware of any existing generalized map for the 50 U.S. states or any of the more than 3,000 counties in the U.S. that satisfies our specific needs, i.e., that is available in electronic format and provides the required level of generalization. Unfortunately, regular maps are unsuitable for use in LM plots, mostly due to two reasons.

First, micromaps in printed form or on a computer display typically are smaller than 2 inches x 2 inches. In this scale, a small region such as Washington, D.C., on the U.S. map would be invisible if drawn at scale. Therefore, micromaps require an exaggeration of small regions such that these regions become visible and can be color-coded.

Second, in an interactive environment such as the Web, the number of line segments determines how fast a new display is drawn and an area is filled with color. The fewer edges a map has, the faster the boundary information is passed from the Web server to the client's computer and the entire graphical display is drawn.

Therefore, it is necessary to develop procedures that are able to create generalized maps with little user interaction. These procedures have to extract selected regions from a larger file and they have to smooth and simplify boundaries by removing details but keeping the topological integrity of a real unit. Micromaps should not end up with holes and neighboring regions in the original map should remain neighboring regions in the generalized map.

The creation of generalized maps for use in the CEP Web site starts with boundary files describing geographical regions and with attribute data describing the characteristics of the regions. These data, the boundary files and attribute data, are often stored in Geographic Information Systems (GIS). In our case, we make use of ArcView, a desktop GIS package, which is one of the most popular GIS. For future use of the developed procedures for map generalization, users need to have access to ArcView and the geographical data must be available in shapefile format. Obviously, the creation of these generalized maps only has to be done once - users of the CEP Web site do not have to deal with this issue.

In the process of creating generalized maps of the U.S. at the sub-state level, we start with a shapefile of the entire U.S. at the preferred level of geography (county, township in the New England states, census tract, or even block group). We assume that in the attribute table each record or areal unit includes a state identifier indicating which state the areal unit belongs to. A procedure written in ArcView's Avenue scripting language allows users to extract the boundary of the selected geographical level (for instance, county) by states to create a shapefile for each state. Thus, each state can be individually displayed.

Most boundary files of the U.S. have a relatively high level of resolution that exceeds the required resolution level for micromaps. As explained earlier, high resolution and detailed data inhibit the fast processing and display of maps on the Web. Therefore, there is a need to smooth or simplify the boundary by removing details but keeping the topological integrity of the areal unit.

A set of Avenue scripts based upon the famous Douglas-Peucker line generalization algorithm (28) has been developed to generalize boundaries to expedite the processing and display of maps on the CEP Web site. The Douglas-Peucker algorithm has been implemented in

many GIS packages (including ARC/INFO) and has been used in numerous occasions. However, the algorithm was designed to generalize linear features such as rivers and roads. It was not intended to generalize polygon features which is the situation in this project. The major problem with using the Douglas-Peucker algorithm to generalize polygon features is to maintain the topological integrity among polygons. This means that neighboring relationships among polygons have to be maintained even after the polygon boundaries have been generalized. The algorithm designed for this project can generalize polygon boundaries and at the same time, it can retain the topological relationships among polygons. The detail of the algorithm is beyond the scope of this paper and will be described and published elsewhere.

The generalization process could be performed before individual maps (by states or by counties) are extracted by the first algorithm or after each state or county file has been created. However, it is desirable to generalize maps by individual states or counties instead of the entire country because boundaries of different states have different levels of cartographic complexity. Thus, different parameter values for the generalization process have to be used to yield desirable generalization results for different states or counties. After boundaries by states or counties have been generalized, the boundaries in ArcView shapefile format are converted into ASCII data in a simple format of polygons depicted by a set of points in latitude and longitude. These coordinates are later used for the micromap displays on the CEP Web site.

4.3 Implementation Issues

The CEP Web site has been designed using numerous common Web formats and styles. While the explanatory pages are mostly based on html files, gif images, and pdf files for larger documents, the pages that provide access to the data are based on JAVA and C code, accessible through Common Gateway Interface (CGI) scripts, and automatically created html and JavaScript documents. Each user interaction that results in a new display (e.g., selecting a new state or a different HAP) invokes such a C-CGI script. This script is a C program that first reads the current parameter settings (representation type, HAP, state, county) from the active Web page and then creates a new html/JavaScript Web page. This new page once again is linked to the same C-CGI script, however different parameters are active on this new Web page.

Other than two C-CGI scripts that are responsible for the top right menu and the lower right data display on the CEP Web site, there exists no hard-coded document that is used for the data display. Each newly visible Web page is created on the fly through the C-CGI scripts.

Developing these two C-CGI scripts required several weeks of programming time. Valuable references during the implementation process were the books by Graham (29) for html, Hoque for JavaScript (30), and Eckel (31) for JAVA. In addition, books by Weinman et al. (32, 33) have been a very good source for general design issues of Web pages. The CGI used for the CEP Web site is based on code developed by Thomas Boutell, freely available on the Web at <http://www.boutell.com/cgic/>.

Two non-statistical Web sites have significantly influenced the design and some of the interactive features of the CEP Web site:

<http://www.usnews.com/usnews/edu/college/corank.htm>,
in particular,

http://www.usnews.com/usnews/edu/college/rankings/natunivs/natu_a.htm,
which allows to interactively sort university rankings according to different criteria, and
<http://www.ran.de>,

in particular,

<http://www.ran.de/cgi/fm/pub/fussball/index>,

which is a nice example how to organize frames and update soccer standings (click on “Bundesliga”, then on “Tabelle”) according to different criteria (by round, home or away, etc.).

It should be noted that the CEP Web site does not use any commercial data base program to access the HAP data. Since the data directly originates from a statistical package that has been used for the modeling, it has not been fed into a data base program first. Instead, a three layer tree-shaped directory structure is used where an individual data file can be directly accessed based on its state, county and census tract FIPS code. Data files have been kept as small as possible to avoid the need for any search in files and therefore accelerate the access to the data. Also, all summary statistics at higher levels (U.S., county) have been pre-calculated to speed up the access to minima, maxima, means, medians, and lower and upper quartiles.

5. Discussion

A first version of the Cumulative Exposure Project Web site went online in March 1998, providing general information and documents related to the project. A major remodeling of the site, providing more extensive descriptions of the air toxics data, took place in November 1998. The release of the data through the CEP Web site was scheduled to begin with interactive tables going online in December 1998. The micromap displays were expected to be posted in the following months. However, EPA ultimately decided not to make the modeled air toxics concentrations available through its public Web site, due to concerns that the estimates for 1990 may not be representative of current conditions. EPA has made files of the air toxics concentrations available to the public by request; data from the study may be more widely disseminated, through the Web as well as other mechanisms, by state and local environmental agencies and by other organizations. Recently, the Environmental Defense Fund (EDF) added data and results based on the Cumulative Exposure Project to their <http://scorecard.org> Web site.

While most work on the CEP Web site was completed as originally planned, micromaps have not been fully integrated within the GPL yet. At the current stage, it seems to be advisable to revise the approach for joining the GPL with micromaps. The BLS GPL was originally developed in 1996, and a new (commercial) version of the GPL has been recently developed by an upcoming software company. This new GPL is currently in its alpha testing phase and it has been scheduled for release in the Summer or Fall of 1999. This new GPL already contains many of the features that were intended for the BLS GPL but were never included in the old version. The inclusion of maps and micromaps based on the generalized maps developed for the CEP Web site appears to be straightforward with the new GPL. Although the BLS GPL is still a useful tool, in particular since its source code can be obtained for free from the Bureau of Labor Statistics, it seems to be advisable to use the new GPL for larger future applications. Several employees in federal agencies such as the Centers for Disease Control and Prevention and the Bureau of Labor Statistics have already expressed interest in investigating the use of the new GPL. Even if the CEP Web site will never be publicly accessible, it seems to be likely that the idea of hierarchical clickable micromaps in the (new) GPL environment might be used in another federal project in the near future.

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URLs

Cumulative Exposure Project (U.S. EPA et al., last update 12/8/98):
<http://www.epa.gov/CumulativeExposure/>

Graphics Production Library (Dan Rope, last update 5/28/97):
http://www.monumental.com/dan_rope/gpl/

Micromaps/S-Plus Code (Daniel B. Carr, last update 12/22/97):
<ftp://galaxy.gmu.edu/pub/dcarr/newsletter/micromap/>

cgic: C library for CGI Programming (Thomas Boutell, last update 1999):
<http://www.boutell.com/cgic/>

1999 College Rankings: US News - edu (U.S. News & World Report Inc., last update 1998):
<http://www.usnews.com/usnews/edu/college/corank.htm>
http://www.usnews.com/usnews/edu/college/rankings/natunivs/natu_a.htm

ran-online (SAT.1 Satelliten-Fernsehen GmbH / IMP - Innovative Medientechnik- und Planungs-GmbH, last update 1999):

<http://www.ran.de>
<http://www.ran.de/cgi/fm/pub/fussball/index>

Scorecard (Environmental Defense Fund, last update 1999):
<http://scorecard.org>

All URLs have been verified in April 1999.

Figure 1: Micromap display at the state level for Pennsylvania for all of its 67 counties. Displayed is the median (with respect to the census tract estimates within each county) of the modeled 1990 benzene concentration in micrograms per cubic meter. Counties are ordered from highest to lowest median benzene concentration.

Figure 2: Tabular display at the state level for Pennsylvania showing the modeled 1990 benzene concentration in micrograms per cubic meter. Displayed are the number of census tracts and summary statistics (e.g., mean and median) with respect to the census tract estimates within each county. Counties are ordered from highest to lowest median benzene concentration. While only 27 of PA's 67 counties are visible in this figure, the remaining 40 counties are accessible through the right scrollbar when looking at this data on the Web.

Figure 3: Tabular display at the county level for Prince George County, Virginia, showing an unusual modeled 1990 acetaldehyde concentration. This observation has been marked as an overestimate after EPA's inspection of the modeled 1990 concentrations. After clicking on the "!" icon, the small explanatory message window displayed at the bottom of this figure pops up.

Figure 1:

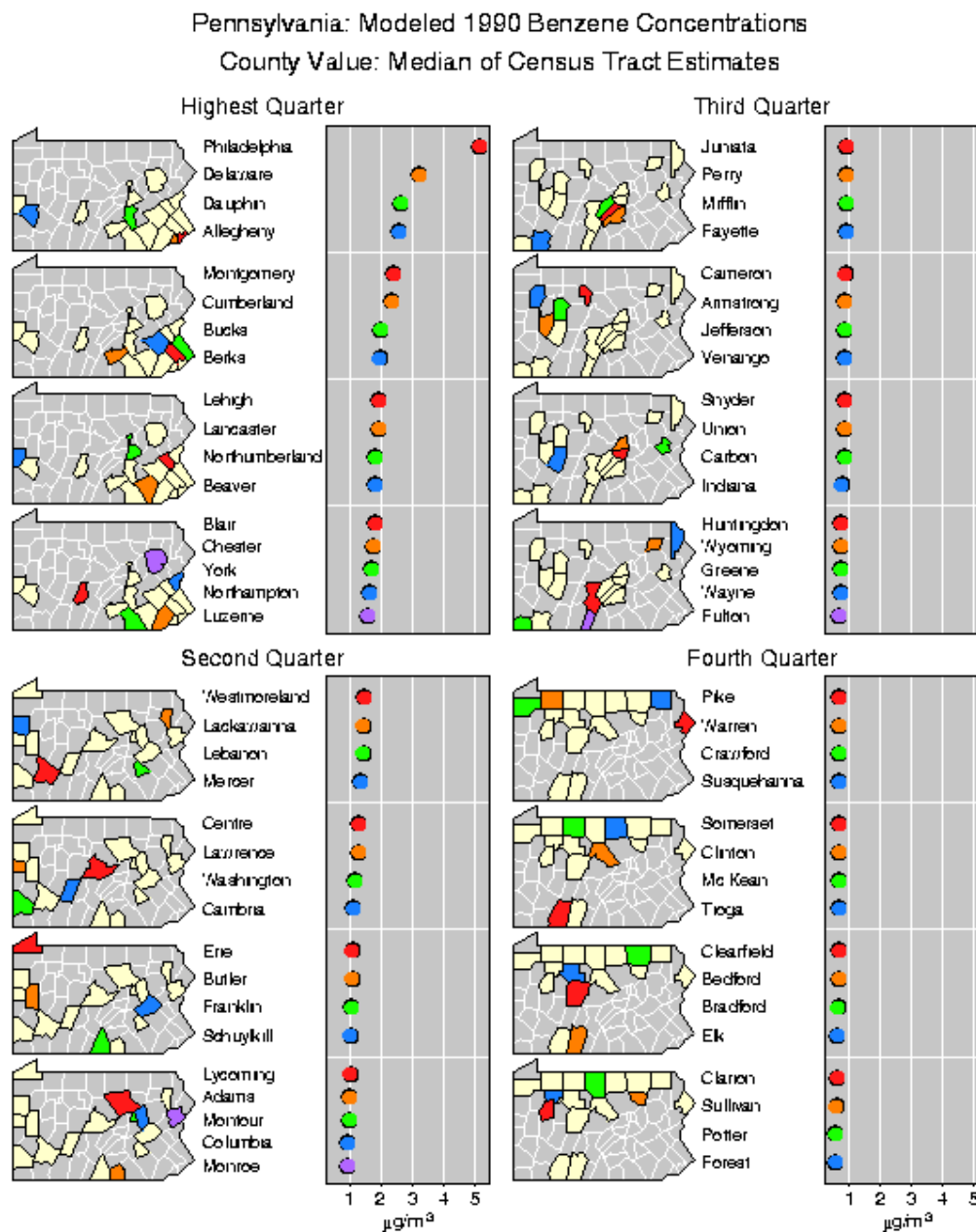


Figure 2:

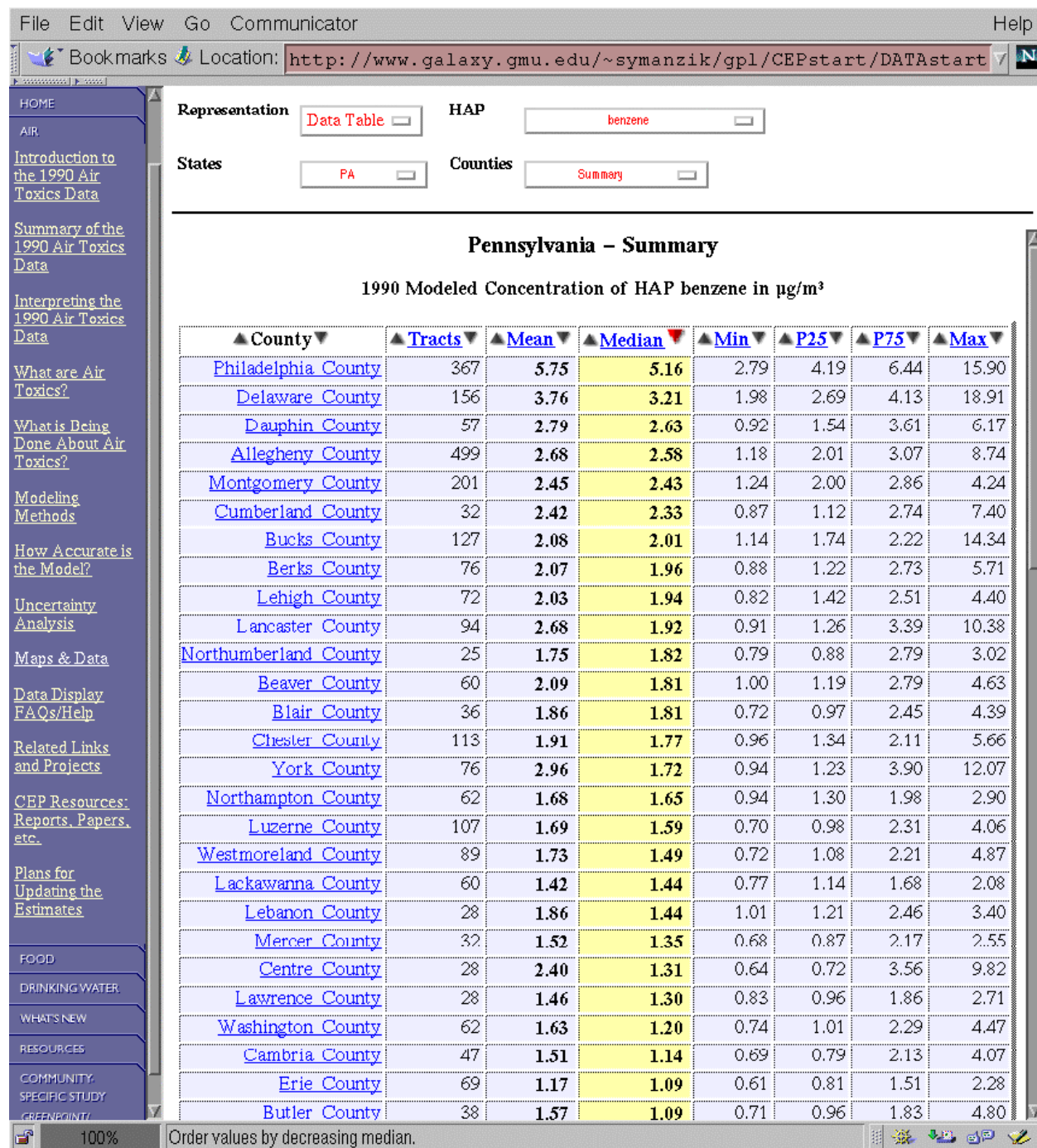


Figure 3:

